DHIRAJ PAL

Python Backend Developer | Django | REST APIs

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PROFESSIONAL SUMMARY

Backend Developer with hands-on experience in Python, Django, FastAPI, Django REST Framework, and PostgreSQL, building REST APIs and database-driven applications for a production water distribution management system (ELINKJAL) and an HRM/ERP system. Skilled in Django ORM, CRUD, authentication, RBAC, SQL, geolocation tracking, and Chart.js visualizations. Built independent projects including a Whisper + vector-embeddings RAG backend and a full-featured Django e-commerce app. Seeking Python Backend Developer / Django Developer roles.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
Backend: Django, Django REST Framework (DRF), FastAPI, REST APIs, Django ORM, CRUD
Databases: PostgreSQL, MySQL, SQLite3, SQL
Authentication & Authorization: Authentication, RBAC, JWT
Frontend: HTML5, CSS3, JavaScript, AJAX, Bootstrap, Chart.js
AI / Data: NumPy, Pandas, OpenAI Whisper, Ollama, Vector Embeddings, Semantic Search, RAG
DevOps: Docker, CI/CD, GitHub Actions, FFmpeg
Tools: Git, GitHub, Postman, VS Code, Jupyter

PROFESSIONAL EXPERIENCE

Elink Software Technologies Pvt. Ltd. *Bandra, Mumbai*
Jr. Python Developer Intern *Oct 2025 – Jan 2026*
Jr. Python Developer (Full-Time) — Promoted from Intern *Feb 2026 – Jun 2026*
Project: ELINKJAL – NRW Water Distribution Management System

- Developed backend functionality and RESTful APIs (DRF) for a water distribution and Non-Revenue Water (NRW) management system using Python, Django, and PostgreSQL, supporting water-meter and flow-rate data.
- Built an engineer management module with secure login/authentication and real-time location tracking, enabling registration and map-based monitoring of 50+ field engineers.
- Built Chart.js-based dashboards to visualize water supply, consumption, and distribution metrics for end users.
- Implemented backend business logic and database operations using Django ORM and PostgreSQL, including CRUD, RBAC, and SQL-based reporting queries for secure, role-managed access.
- Integrated JavaScript/AJAX for asynchronous data loading, and used Git/GitHub for version control and team collaboration while debugging backend issues.

Sky Need Solution Pvt. Ltd.
Jr. Backend Developer Intern *Oct 2023 – Mar 2024*
Project: ERP – HRM Management System

- Developed backend modules for an HRM system using FastAPI and PostgreSQL, covering employee management, attendance tracking, and leave management, including CRUD APIs for employee records.
- Developed a leave request and approval workflow with SMTP-based email notifications, and implemented authentication to secure HRM API endpoints.

PROJECTS

RAG-Based AI Chatbot – Backend Application *Jul 2026 – Present*
Python, FastAPI, Whisper, Ollama, RAG, Embeddings, Vector Search | GitHub

- Built a Retrieval-Augmented Generation (RAG) backend pipeline in Python: automated video-to-audio conversion (FFmpeg) and transcribed/translated Hindi audio using OpenAI's Whisper (large-v2), processing 4+ videos totaling over 2 hours of content.
- Implemented timestamped text chunking with structured JSON storage, generated semantic vector embeddings using Ollama's bge-m3 model, and integrated retrieval with a local LLM for context-aware responses — a backend-focused RAG system without a separate frontend UI.

Shopee – Django E-Commerce Web Application *Dec 2024 – Apr 2025*
Python, Django, SQLite3 | GitHub

- Developed a Django-based e-commerce application with user authentication, product catalog (10+ products across multiple categories), shopping cart, checkout, and order management.
- Implemented add-to-cart functionality along with category-based filtering and product sorting to improve product discovery.
- Implemented backend logic using Django ORM and SQLite3, built an admin dashboard with CRUD for products/categories/users/orders, and designed responsive frontend pages using HTML5, CSS3, and Bootstrap.

EDUCATION

Nirmala Memorial Foundation College, Mumbai *Mumbai, Maharashtra*
Bachelor of Science in Information Technology (B.Sc. IT) *Aug 2021 – May 2025*
CGPA: 7.0 / 10